SYSTEM ARCHITECTURE / SGLANG V0.5.9

Deterministic Decode: opt-in GEMM specialization

The performance branch adapts the small-M GEMM path; native scheduling and CUDA Graph eligibility remain authoritative.

SERVING PATH

01 Request batching
SGLang Scheduler
fixed serving workload

→

02 ModelRunner
decode forward
native eligibility check

→

03 CUDA Graph path
capture / replay
no recapture introduced

→

04 Model outputs
greedy token stream
checked request by request

INSIDE THE PERSISTENT TRITON MATMUL

OPT-IN GATE
dtype = BF16 M <= 16
K % 64 = 0 N % 128 = 0
compatible strides / explicit switch

CANDIDATE
M16 / N64 / K128 · 4 warps

FALLBACK
native persistent Triton GEMM

SHARED NUMERICAL CONTRACT
FP32 accumulation and bias
output shape / dtype preserved
exact-output validation

VALIDATION EVIDENCE

TRACE
Execution-path identity
63 CUDA Graph launches / arm

NUMERICS
Output equivalence
340 GEMM + 255 reorder checks

PERFORMANCE
Paired serving A/B
5 pairs · median +72.4%